A short note on three elementary inequalities

Benchmark manuscript for the TeX proof-check kit

Abstract

We record three elementary inequalities for real numbers that are used repeatedly in later work. The note is self-contained except for one classical inequality, which we quote from the literature.

1 Notation

Throughout, $\mathbb{R}$ denotes the real numbers and $\mathbb{R}_{>0}$ the positive real numbers. For real numbers x, y, z we write

$$S(x,y,z) = \frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}$$

whenever the three denominators are nonzero.

2 Results

Lemma 2.1. *For all real numbers a and b we have*

$$ab \le \frac{a^2+b^2}{2},$$

with equality if and only if $a = b$.

Proof. Since $(a-b)^2 \ge 0$, expanding the square gives $a^2 - 2ab + b^2 \ge 0$, that is, $2ab \le a^2 + b^2$. Dividing by 2 gives the inequality. Moreover, equality holds if and only if $(a-b)^2 = 0$, that is, if and only if $a = b$. □

Proposition 2.2. *For all $x, y, z \in \mathbb{R}_{>0}$ we have*

$$S(x,y,z) \ge \frac{3\,(xy+yz+zx)}{2\,(x^2+y^2+z^2)}.$$

Proof. By Nesbitt's inequality [1, Theorem 1], $S(x,y,z) \geq 3/2$ for all $x, y, z \in \mathbb{R}_{>0}$. Applying Lemma 2.1 to the pairs (x,y), (y,z) and (z,x) and adding the three inequalities gives

$$xy + yz + zx \leq x^2 + y^2 + z^2.$$

Since $x^2 + y^2 + z^2 > 0$, dividing by it yields $\frac{xy+yz+zx}{x^2+y^2+z^2} \leq 1$, and hence $\frac{3\,(xy+yz+zx)}{2\,(x^2+y^2+z^2)} \leq \frac{3}{2} \leq S(x,y,z)$. □

Lemma 2.3. *For every real number $x \neq -1$ we have*

$$\frac{x^2}{1+x} \geq x - 1.$$

Proof. Multiplying both sides by $1 + x$, the claim is equivalent to $x^2 \geq (x-1)(x+1) = x^2 - 1$, which is obvious. □

Remark 2.4. Lemma 2.3 is used in the companion paper only for $x \geq 0$.

References

[1] A. M. Nesbitt, Problem 15114, *Educational Times* **55** (1903), 37.